

# Don Santos

[don16santos@gmail.com](mailto:don16santos@gmail.com) • +27 67 921 5702 • Gauteng, South Africa • [LinkedIn](#) • [GitHub](#)

## PROFESSIONAL SUMMARY

Systems Architect and Full-Stack Engineer specializing in designing fault-tolerant AI orchestration pipelines, distributed multi-agent swarms, and high-performance React architectures. Expert in resolving the compounding error rates of single-instance large language models through system-level redundancy, consensus-based voting protocols, and recursive task decomposition. Partnering with startups and founders to build production-grade, deterministic AI platforms and user interfaces that maximize throughput and visual excellence.

## PROFESSIONAL EXPERIENCE

### Full-Stack Architect & Creator | ResumAI

- Architected and built ResumAI, an industrial-grade CV builder and job application ecosystem designed to replace unstable, single-agent AI text generation with a 5-worker consensus Swarm Engine.
- Designed and implemented Maximal Agentic Decomposition protocols, splitting complex document parsing and optimization workflows into granular, low-token decisions to mathematically suppress generation errors.
- Created a Gold Standard PDF rendering pipeline leveraging headless Puppeteer and custom print stylesheets, ensuring pixel-perfect, ATS-optimized, Harvard-style PDF generation.
- Engineered a multi-threaded batch application workflow ("Mass Apply") that tails job applications, dynamically customizes resumes, and manages batch submissions with redundant worker verification.
- Developed a real-time Quality Intelligence service providing semantic scoring, metric-driven text suggestions, and resume enhancement feedback using local LLM models.

## EDUCATION

### Diploma in Paralegal Studies | University of Johannesburg (UJ)

### Full-Stack Certificate in Web Development | HyperionDev

### Ongoing Research in Distributed Systems & LLM Error Correction | Independent Engineering Research

## SKILLS

**AI & Systems Design:** Swarm Intelligence, Ensemble Voting, Maximal Decomposition, RAG Architectures, Vector Databases, LangChain, Ollama / Groq

**Frontend Engineering:** React, Next.js 14, TypeScript, Tailwind CSS, Framer Motion, WebGL Shaders

**Backend & Tooling:** Node.js / Express, REST & WebSockets, SQLite / Postgres, Puppeteer Rendering, Docker, Git / CI-CD

## PROJECTS

### DocketDive — AI Legal Research Assistant

[View Project](#)

South Africa's first AI-native legal research assistant. Created a multi-stage Retrieval-Augmented Generation (RAG) system that allows legal professionals to query, analyze, and cite case law with zero hallucinations.

### StudyPodLM — Multi-Modal Learning Workspace

[View Project](#)

An interactive AI education platform converting textbooks, lecture notes, and video assets into structured study guides, complete with custom text-to-speech rendering and self-assessment engines.

### Design Systems Lab — High-Fidelity UI Playground

[View Project](#)

An experimental showcase for cutting-edge user interfaces, containing complex WebGL shaders, fluid dynamics, glassmorphism components, and spring-physics micro-animations.